

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. *(BL4)*

Assessment Tool Weightage: 40%

Code of Conduct:

*I **TEJASWINI M (192411079)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

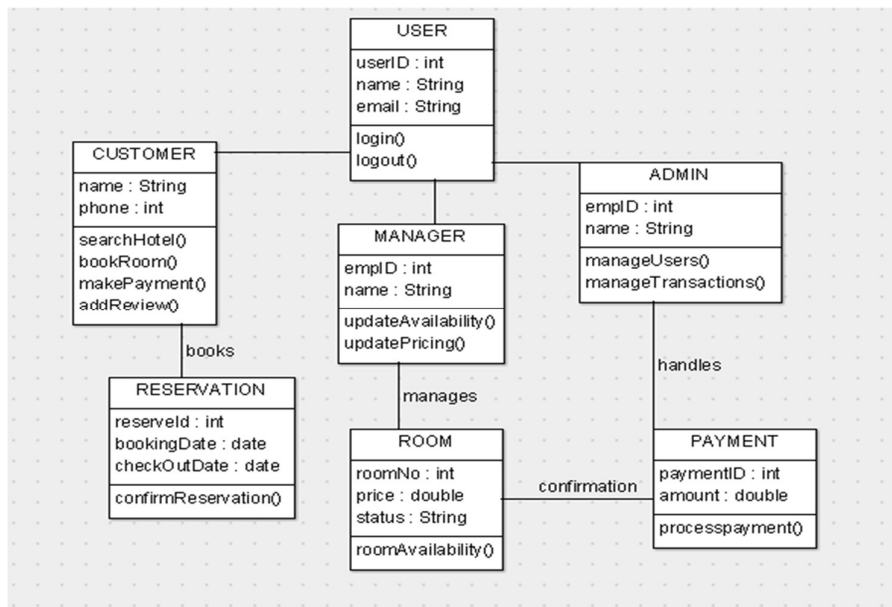
TEJASWINI M

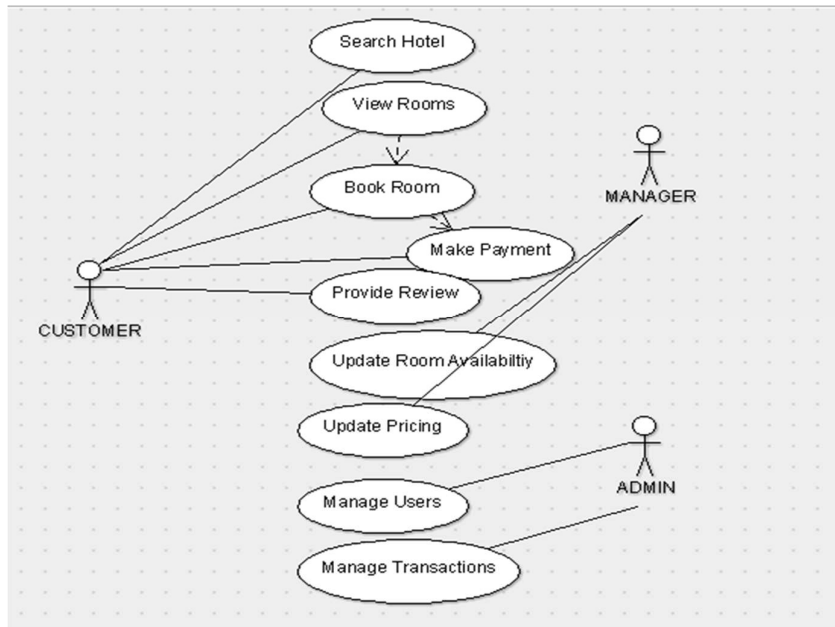
1. Online Hotel Reservation System

A travel company plans to develop an online hotel reservation platform where customers can search hotels, book rooms, make payments, and provide reviews. Hotel managers can update room availability and pricing, while administrators manage users and transactions.

Question:

Analyze the system and explain how object-oriented principles (encapsulation, inheritance, polymorphism) can be applied. Also, suggest a suitable SDLC model and justify your choice for this system.





Suitable SDLC Model

Agile Model

The **Agile SDLC model** is the most suitable choice for an Online Hotel Reservation System.

Reasons

- Customer requirements may change frequently.
- New features such as **discount offers, cancellation policies, loyalty programs, ratings, and additional payment methods** can be added incrementally.
- Continuous feedback from customers and hotel managers is important.
- Agile supports **iterative development, testing, and deployment**, which helps deliver working features quickly.

Advantages of Agile for This System

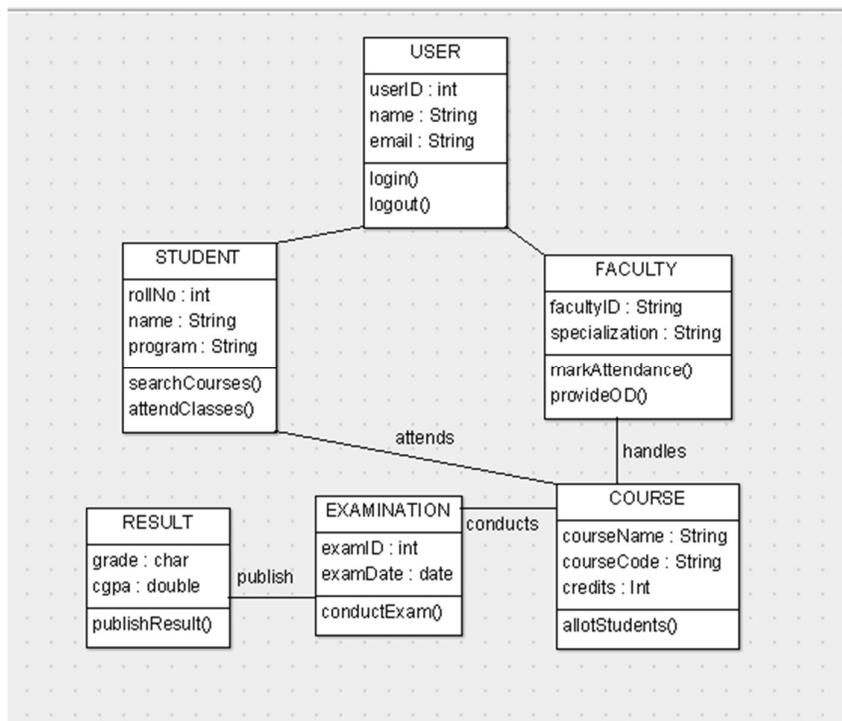
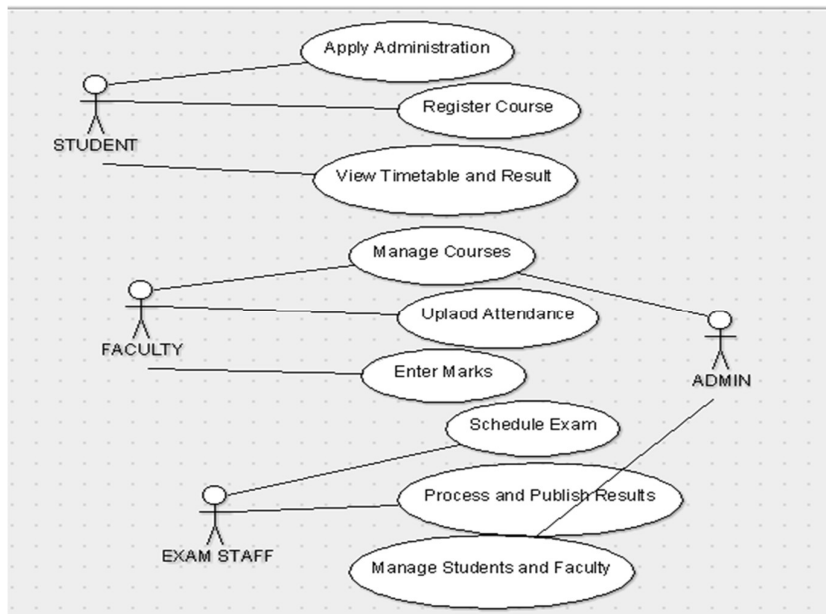
- Faster development and delivery.
- Easy handling of changing requirements.
- Improved software quality through continuous testing.
- Better customer satisfaction because feedback is incorporated regularly.

2. University Management System

A university intends to automate its operations including student admissions, course registration, examination management, and result processing. Different users such as students, faculty members, examination staff, and administrators interact with the system.

Question:

Design a structured approach using object-oriented concepts to model this system. Identify key classes and relationships, and recommend an SDLC model for efficient development with reasoning.



Recommended SDLC: Agile Model

Why Agile?

The university system contains many modules:

- Admissions
- Course Registration
- Examination Management
- Result Processing
- Reports and Analytics

These modules can be developed incrementally.

Reasoning

- Requirements may change every semester.
- New regulations and examination rules may be added.

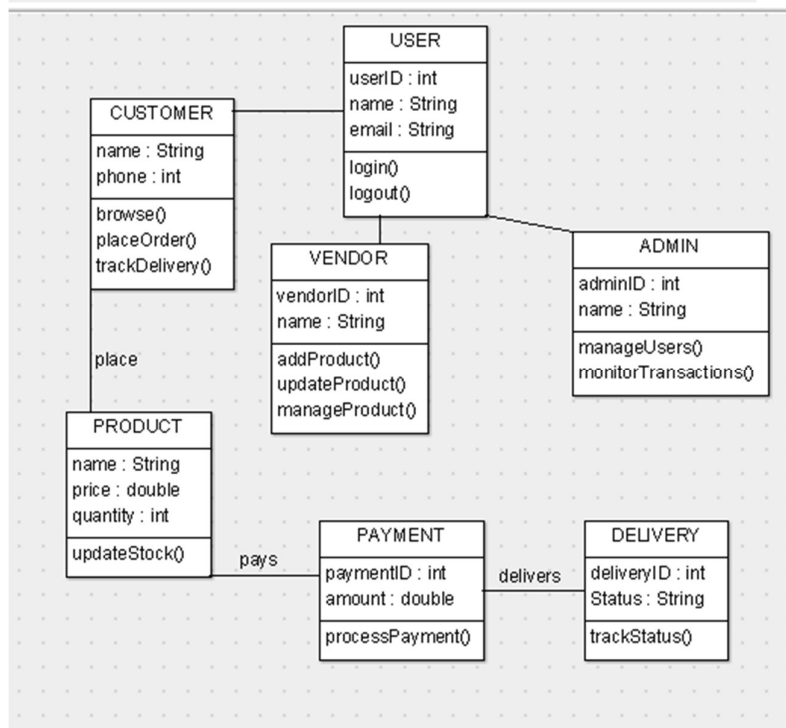
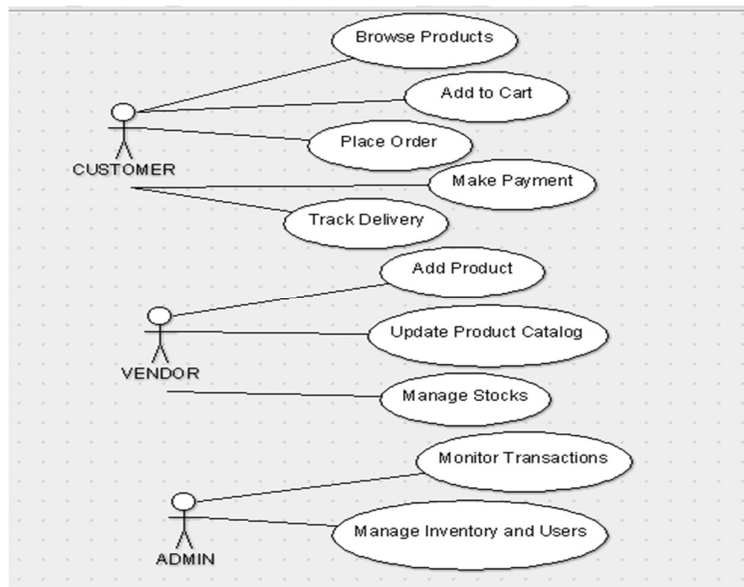
- Different departments may request modifications.
- Agile allows continuous testing and feedback from students, faculty, and administrators.

3. Online Retail Shopping Platform

A retail company wants to develop an e-commerce platform where customers can browse products, place orders, track deliveries, and make online payments. Vendors manage product catalogs, while administrators monitor transactions and inventory.

Question:

Discuss how modular software design can be achieved using object-oriented techniques. Evaluate different SDLC models and select the most appropriate one for this scenario with justification.



Object-Oriented Concepts Applied

Encapsulation

Sensitive data such as price, stock, and payment amount is stored inside classes and accessed through methods like `updateStock()` and `processPayment()`.

Inheritance

The Customer, Vendor, and Administrator classes inherit common properties from the User class, reducing code duplication.

Polymorphism

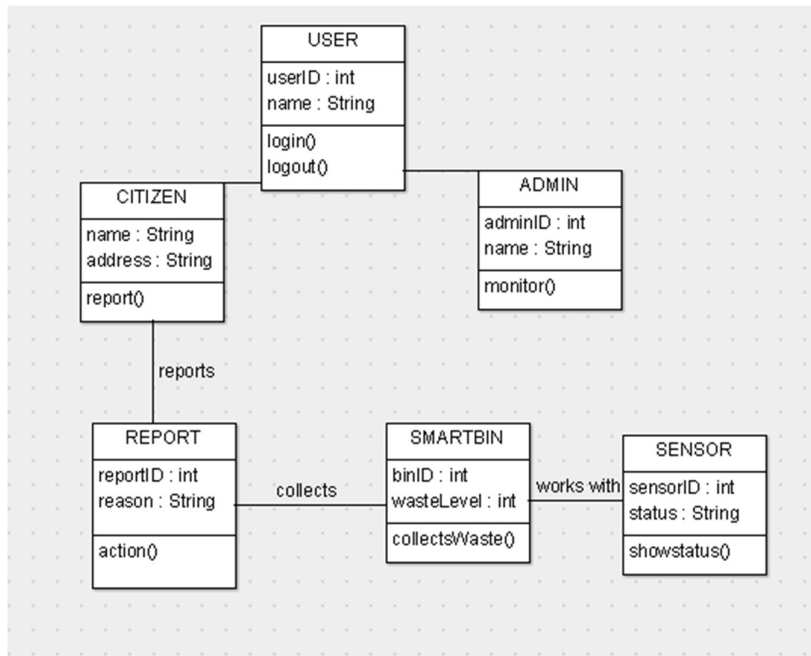
A common operation such as `accessDashboard()` can behave differently for customers, vendors, and administrators, providing **flexibility** in the system.

4. Smart Waste Management System

A municipality introduces a smart waste management system where IoT-enabled bins monitor waste levels and automatically notify collection vehicles when bins are full. Citizens can also report waste-related issues through a mobile application.

Question:

Apply object-oriented system development principles to design this system. Identify the life cycle model best suited for handling real-time data, sensor integration, and evolving requirements, and explain why.



Object-Oriented Principles Applied

Encapsulation

Each class keeps its data private.

Example:

- SmartBin stores `binId`, `wasteLevel`, and location.
- Waste levels can be updated only through methods such as `updateLevel()`.

This improves data security and integrity.

Inheritance

A common superclass `User` can be created.

Subclasses:

- Citizen
- Administrator
- VehicleOperator

This reduces code duplication and supports reusability.

Polymorphism

A common method such as `sendNotification()` can behave differently for:

- citizens,
- vehicle operators,
- administrators.

This provides flexibility in the notification system.

Best Life Cycle Model

Recommended Model: Agile + DevOps

This system involves:

- real-time IoT sensor data,
- cloud communication,
- mobile applications,
- vehicle tracking,
- continuous updates,
- and evolving municipal requirements.

A combination of Agile and DevOps is the most suitable approach.

Why Agile?

Agile supports:

- iterative development,
- frequent requirement changes,
- continuous user feedback,
- and incremental delivery of modules such as:
 - Smart Bin Monitoring,
 - Mobile Reporting,
 - Vehicle Notification,
 - Analytics Dashboard.

Why DevOps?

DevOps is important because the system must operate continuously in real time.

It supports:

- automated testing,
- continuous integration,
- continuous deployment,
- cloud monitoring,
- and rapid issue resolution.

This is especially useful for IoT sensor integration and live municipal operations.

Comparison of SDLC Models

Model	Suitability
-------	-------------

Waterfall	Poor – difficult to handle changing requirements
Prototype	Good for designing the mobile app interface
Spiral	Good for risk management but more complex
Agile	Excellent for evolving functional requirements
DevOps	Excellent for real-time deployment and monitoring

Why Agile + DevOps Is the Best Choice

Advantages

- Supports real-time sensor integration
- Allows frequent feature updates
- Handles changing municipal policies
- Enables continuous deployment of IoT services
- Improves system reliability and scalability
- Reduces downtime for critical waste collection operations